

- 1(a) As part of a research project, the masses, m grams, of a random sample of 1000 pebbles from a certain beach were recorded. The results are summarised in the table.

Mass (g)	$50 \leq m < 150$	$150 \leq m < 200$	$200 \leq m < 250$	$250 \leq m < 350$
Frequency	162	318	355	165

Calculate estimates of the mean and standard deviation of these masses.

[2]

- (b) The masses, x grams, of a random sample of 1000 pebbles on a different beach were also found. It was proposed that the distribution of these masses should be modelled by the random variable $X \sim N(200, 3600)$.

Use the model to find $P(150 < X < 210)$.

[1]

- (c) Use the model to determine x_1 such that $P(160 < X < x_1) = 0.6$, giving your answer correct to **five** significant figures. [3]

- (d) It was found that the smallest and largest masses of the pebbles in this second sample were 112 g and 288 g respectively.

Use these results to show that the model may not be appropriate. [1]

- (e) Suggest a different value of a parameter of the model in the light of these results. [2]

2 The masses, in grams, of plums of a certain kind have the distribution $N(55, 18)$.

(i) Find the probability that a plum chosen at random has a mass between 50.0 and 60.0 grams.

[1]

(ii) The heaviest 5% of plums are classified as extra large.

[1]

Find the minimum mass of extra large plums.

(iii) The plums are packed in bags, each containing 10 randomly selected plums.

Find the probability that a bag chosen at random has a total mass of less than 530 g.

[4]

3 A fair dice is thrown 1000 times and the number, X , of throws on which the score is 6 is noted.

(a) (i) State the distribution of X . [1]

(ii) Explain why a normal distribution would be an appropriate approximation to the distribution of X . [1]

(b) Use a normal distribution to find two positive integer values, a and b , such that $P(a < X < b) \approx 0.4$. [5]

- (c) For your two values of a and b , use the distribution of part (a)(i) to find the value of $P(a < X < b)$, correct to 3 significant figures. [2]

- 4 The masses, X grams, of tomatoes are normally distributed. Half of the tomatoes have masses greater than 56.0 g and 70% of the tomatoes have masses greater than 53.0 g.

- (a) Find the percentage of tomatoes with masses greater than 59.0 g. [2]

- (b) Find the percentage of tomatoes with masses greater than 65.0 g. [4]

(c) Given that $P(a < X < 50) = 0.1$, find a .

[3]

- 5 The heaviest 17% of rococo apples are classified as large, and the lightest 17% are classified as small. The remainder are classified as medium. The limits within which the masses of medium rococo apples lie are 96 g and 120 g. Stating a necessary assumption, estimate the mass of the heaviest rococo apple. [4]

6 The variable X has the distribution $N(20, 4^2)$.

(a) Given that $P(X < a) = 0.1$, find a .

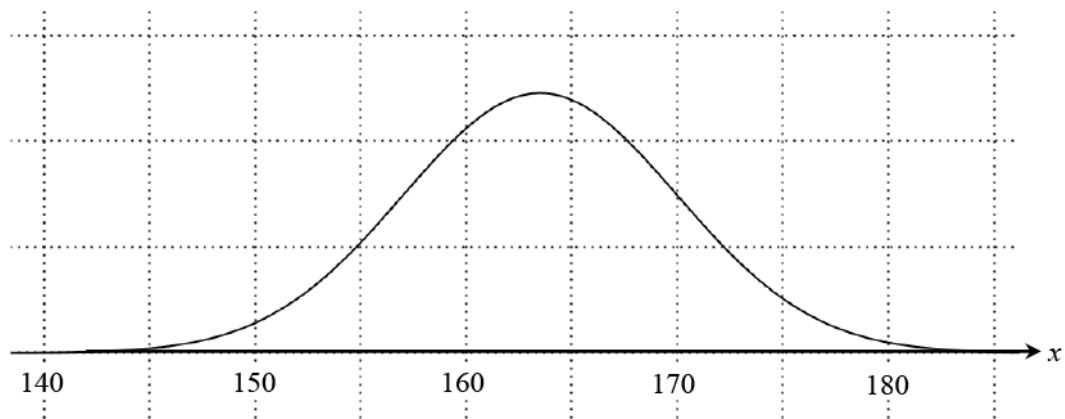
[1]

(b) Given that $P(b < X < c) = 0.95$, find a possible pair of values of b and c .

[2]

- 7 (a) The heights of English men aged 25 to 34 are normally distributed with mean 178 cm and standard deviation 8 cm. Three English men aged 25 to 34 are chosen at random. Find the probability that all three of them have a height less than 194 cm. [3]

- (b) The diagram shows the distribution of heights of Scottish women aged 25 to 34.



It is given that the distribution is approximately normal. Use the diagram above to estimate the standard deviation of these heights, explaining your method. [3]

- 8 A market gardener records the masses of a random sample of 100 of this year's crop of plums. The table shows his results.

Mass, m grams	$m < 25$	$25 \leq m < 35$	$35 \leq m < 45$	$45 \leq m < 55$	$55 \leq m < 65$	$65 \leq m < 75$	$m \geq 75$
Number of plums	0	3	29	36	30	2	0

- (a) Explain why the normal distribution might be a reasonable model for this distribution. [1]

The market gardener models the distribution of masses by $N(47.5, 10^2)$.

- (b) Find the number of plums in the sample that this model would predict to have masses in the range
(i) $35 \leq m < 45$, [2]

- (ii) $m < 25$. [2]

- (c) Use your answers to parts (b)(i) and (b)(ii) to comment on the suitability of this model. [1]

- (d) The market gardener plans to use this model to predict the distribution of the masses of next year's crop of plums. Comment on this plan. [1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		(mean =) 201 (3 sf) (sd =) 60.7 (3 sf)	B1 (AO1.1) B1 (AO1.1) [2]	Allow 60.8
	b		0.364 (3 sf)	B1 (AO3.4) [1]	
	c		$P(X < 160) = 0.252(49)$ $x_1 = \Phi^{-1}(0.6 + '0.25249')$ $= 262.83$ (5 sf) ISW	B1 (AO3.4) M1 (AO1.1) A1 (AO1.1) [3]	soi, eg by $P(X > 160) = 0.748$ or 0.747 or by 0.147 or 0.148 T&I: correct answer scores B1M1A1, otherwise max B1 SC Answer 263 with correct working: B1M1A0 SC Answer 263 with inadequate working: B1 only
	d		112 and 288 are within 2 sd from mean (no working needed) $P(X < 112) = 0.0708$, which is > 0.025 or > 0.0013 or > 0	B1 (AO3.5a) [1]	or $\mu + 2\sigma = 320$ ($\mu + 3\sigma = 380$) which is $>$ than 288 or $P(112 < M < 288) = 0.858$ which is $<$ than 0.95 (or 0.99) or $p = 0.858$, but model suggests $p = 1$ NOT 0.858 alone B0
	e		Reduce σ $288 - 200 = 2\sigma$ or $288 - 200 = 3\sigma$ or $288 - 112 = 4\sigma$ or $288 - 112 = 6\sigma$ $\sigma = 44$ $\sigma = 29.3$ or about 30	B1 (AO3.5c) B1 (AO3.3) [2]	May be implied by value of σ Allow more precise correct methods Allow σ between 25 and 50. No working needed B1B1 or σ^2 between 625 and 2500 <u>Examiner's Comments</u> This question is best answered by noting that most (or almost all) the values of a normal distribution lie within 2 (or 3) standard deviations of the mean. However, many candidates who found these values did not explicitly compare them with the two values given in the question.
			Total	9	
2		i	0.761 or 0.762 (3 sf)	B1(AO1.1) [1]	BC Allow 0.76 <u>Examiner's Comments</u> This question was well answered.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
		ii	62.0 (3 sf)	B1(AO1.1) [1]	BC Allow 62 or 61.9 <u>Examiner's Comments</u> This question was also well answered.	Allow $m \geq 62.0$
		iii	Use of $\bar{X}$ eg " $\bar{X}$ " or "mean" or $\frac{18}{10}$ or $\sqrt{\frac{18}{10}}$ $\bar{X} \sim N(55, N(55, \frac{18}{10}))$ $P(\bar{X} < \frac{530}{10} \text{ dep } \sigma^2 = \frac{18}{10})$ = 0.0680 (3 sf)	M1(AO1.1a) M1(AO3.3) M1(AO3.4) A1(AO1.1) [4]	$\mu = 550$ seen or implied $\Sigma \bar{X} \sim N(550, 180)$ Correct $P(\Sigma \bar{X} < 530) \text{ dep } \sigma^2 = 180$ = 0.0680 (3 sf) Allow 0.068	May be implied Stated or implied Correct answer from limited (or no) working: M1M1M1A1 <u>Examiner's Comments</u> A disappointingly small number of candidates recognised that this question could be answered using the distribution of $\bar{X}$, the sample mean. Those who did recognise this were generally successful. A few candidates used the formulae (not in the specification, but acceptable) $E(X_1 + X_2 + X_3 + \dots + X_n) = nE(X)$ and $\text{Var}(X_1 + X_2 + X_3 + \dots + X_n) = n\text{Var}(X)$, and answered correctly. However, some used $n^2\text{Var}(X)$, and so obtained a standard deviation of $\sqrt{1800}$ instead of $\sqrt{180}$. Many assumed that each plum had to have a mass of less than 53 g. These found $(P(X=53))^{10}$.
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
3		a	(i) X is binomial (ii) Large n	B1 (AO 3.3) [1] B1 (AO 3.3) [1]		
		b	$X \sim N(\frac{500}{3}, \frac{1250}{9})$ e.g. $P(X < b) = 0.7$ $b = 173$ or 174 $a = \frac{500}{3} - ((173 \text{ or } 174) - \frac{500}{3})$ $= 160$ or 159	B1 (AO 1.2) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 1.1) [5]	soi; allow $N(167, 139)$ BC or $P(X < a) = 0.3$	Other correct methods score similarly eg $\Phi^{-1}(0.9)$ $= 181$ $\Phi^{-1}(0.5)$ $= 166$
		c	$X \sim \text{Bin}(1000, \frac{1}{6})$ eg $P(160 < X < 173)$ or $P(159 < X < 174)$ $= 0.69218 - 0.30280$ or $0.72108 - 0.27355$ $= 0.389$ (3 sf) or 0.448 (3 sf)	M1 (AO 3.4) A1 (AO 1.1) [2]	BC NB ft their a and b	$P(166 < X < 181)$ $0.87910 - 0.49812$ $= 0.381$ (3 sf)
			Total	9		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		a	$\mu = 56$ Percentage with masses > 59 g = 30%	B1(AO 1.1a) B1(AO 1.1) [2]	or 0.3
		b	$\Phi\left(\frac{53-56}{\sigma}\right) = 0.3, \frac{53-56}{\sigma} = -0.5244$ $\sigma = 5.721$ $X \sim N(56, '5.721^2)$ soi $P(X > 65) = 0.0578$ or 5.78% (3 sf)	M1(AO 2.1) A1(AO 1.1) M1(AO 2.4) A1(AO 1.1) [4]	or $P(X > 65) = P(z > \frac{65-56}{5.721})$ ft their σ $= P(z > 1.573)$ Or BC
		c	$P(X < 50) = 0.1471$ $P(X < a) = 0.0471$ $a = 46.4$ (3 sf)	M1(AO 1.1a) A1(AO 2.1) A1(AO 1.1) [3]	
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5			Assume masses normally distr. 66% of masses lie approximately within $\mu \pm \sigma$ and greatest mass $\approx \mu \pm 3\sigma$ sd = $0.5(120 - 96)$ (= 12) Greatest mass = $120 + 2 \times 12 = 144$ (g)	B1(AO 1.2) M1(AO 3.1b) M1(AO 1.1) A1(AO 1.1) [4]	or similar both stated or implied or greatest mass $= \frac{96+120}{2} + 3 \times 12 = 144$	Allow 144 to 145
			Total	4		
6		a	14.9 (3 sf)	B1(AO 1.1) [1]	BC	
		b	0.975 seen or implied $c = 27.8$, $b = 12.2$ (3 sf)	M1(AO 1.1a) A1(AO 1.1) [2]	Other solutions are possible	
			Total	3		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
7		a	$N(178, 8^2)$ and $X < 194$ oe $P(X < 194) = 0.977(249868\dots)$ $0.977249868\dots^3 = 0.933$ (3 s.f.)	M1(AO1.1) A1(AO1.1) A1(AO1.1) [3]	Soi BC	
		b	E.g. inflection – mean E.g. $\frac{1}{2}$ (97.5th percentile – mean) E.g. $\frac{1}{6}$ (99.7th percentile – 0.3th percentile) = 6 to 7 E.g. Point of inflection is 1 sd from mean E.g. 95% of values within (approx) 2 sds of mean E.g. Almost all within (approx) 3 sds of mean	M1(AO1.1a) A1(AO1.1) E1(AO2.4) [3]	E.g. 170 – 163 E.g. $\frac{1}{2}$ (176 – 163) E.g. $\frac{1}{6}$ (183 – 145) Statement matching method used	Figures are illustrative only
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
8		a	Symmetrical, high in middle, tails off at ends	B1(AO2.4) [1]	Any two of these	Not just bell shaped
		b	(a) $P(35 < m < 45) = 0.296$ Predicted no. = 30	M1(AO3.4) A1(AO1.1) [2]	Correct probability attempted Allow 29.6 or '29 or 30'	
		c	(b) $P(m < 25) = 0.0122$ Predicted no. = 1	M1(AO3.4) A1(AO1.1) [2]	Correct probability attempted Allow 1.2 or '1 or 2'	
		d	29.6 close to 29 and 1.2 close to 0 Hence model (could be) suitable	B1(AO3.5a) [1]	Both needed	OR B1 Model predicts some masses below 25 g, hence not suitable
		e	E.g. Weather may cause different distribution	B1(AO3.5b) [1]	Any sensible reason why next year may be different	
			Total	7		